

N+ Network Security Assessment

Theme: Startup Office IT Setup

Section A – Scenario-Based Concept Application

Question 1. Explain the selection process for CPU, RAM, and storage specifications to ensure hardware compatibility and prevent performance bottlenecks for a startup team performing concurrent development and video conferencing tasks.

Answer. Selecting the right hardware is essential for a startup of ice where employees perform software development, multitasking, and video conferencing. A balanced combination of CPU, RAM, and storage improves performance and prevents system bottlenecks.

1. CPU Selection

The CPU is responsible for processing all instructions and managing system performance.

Recommended Specifications:

- Intel Core i5/i7 (12th or 13th Generation) or AMD Ryen 5/7
- Minimum 6 Cores and 12 Threads
- Clock speed above 2.5 GHz
- Support for virtualization (Intel VT-x/AMD-V) Benefits:
- Faster processing
- Smooth multitasking
- Better video conferencing
- Faster software compilation

2. RAM Selection

RAM temporarily stores data used by running applications. Recommended Specifications:

- Minimum 16 GB DDR4 RAM
- Recommended 32 GB for heavy workloads
- Dual-channel memory configuration Benefits:
- Faster application switching
- Better multitasking
- Reduced system lag
- Improved virtual machine performance

3. Storage Selection

A high-speed SSD significantly improves system performance.

Storage Type	Recommended Capacity	Purpose
NVMe SSD	512 GB – 1 TB	Operating System & Applications
HDD (Optional)	1–2 TB	Backup & Data Storage

4. Hardware Compatibility

Before purchasing

- CPU socket matches the motherboard.
- RAM type and speed are supported.
- SSD interface (NVMe/SATA) is compatible.
- Power Supply Unit provides sufficient wattage.

components, ensure:

5. Preventing Performance Bottlenecks Common bottlenecks include:

- Insufficient RAM
 - Slow HDD instead of SSD
 - Outdated CPU
 - Poor cooling system
 - Low-capacity power supply

Recommended Configuration

Component	Specification
CPU	Intel Core i5 / AMD Ryzen 5
RAM	16 GB DDR4 (32 GB Recommended)
Storage	512 GB NVMe SSD
Operating System	Windows 11 Pro

Conclusion

Choosing a modern CPU, sufficient RAM, and fast SSD storage ensures smooth multitasking, reliable performance, and efficient productivity while preventing hardware bottlenecks.

Question 2. Explain how system resource allocation and hardware bottlenecks contribute to performance degradation after the deployment of resource-intensive background applications like antivirus software.

Answer. Resource-intensive background applications such as antivirus software continuously monitor files, memory, and network activity to protect the system. While they improve security, they also consume CPU, RAM, storage, and network resources, which may reduce overall system performance if hardware resources are limited.

1. CPU Usage

Antivirus software performs real-time scanning and background processing, increasing CPU utilization.

Effects:

- Slow application response
- Delayed processing
- Increased system temperature

2. RAM Usage

Background services occupy system memory.

When RAM is insufficient:

- Windows uses virtual memory
- Applications become slower
- Multitasking performance decreases

3. Storage Bottleneck

Continuous file scanning increases disk read/write operations.

Effects:

- Slow boot time
- Delayed application loading
- Reduced SSD/HDD performance

4. Network Usage

Automatic updates and traffic monitoring consume internet bandwidth.

Effects:

- Slow internet speed
- Video conferencing interruptions
- Increased network latency

Performance Symptoms

- High CPU usage
- High RAM consumption
- Slow startup
- Frequent disk activity □ System lag
- Reduced productivity Methods to Improve Performance □ Upgrade RAM to 16 GB or more.
- Replace HDD with NVMe SSD.
- Schedule antivirus scans during idle hours.
- Disable unnecessary startup applications.
- Keep antivirus software updated.

Conclusion

Proper resource allocation and hardware upgrades help reduce performance degradation while maintaining effective system security.

Question 3. Explain the systematic troubleshooting process for resolving peripheral hardware failures caused by driver incompatibility or operating system conflicts when a device functions inconsistently across different workstations.

Answer. Peripheral devices such as printers, scanners, keyboards, webcams, and USB devices may work on one computer but fail on another because of driver incompatibility, operating system conflicts, or incorrect hardware settings. A systematic troubleshooting approach helps identify and resolve the issue efficiently.

Troubleshooting Steps

Step 1: Check Physical Connections □

Verify cables and power supply.

- Test different USB ports.
- Replace faulty cables if necessary.

Step 2: Test on Another Computer

- Connect the device to another workstation.
- If it works, the problem is likely with the original computer.

Step 3: Check Device Manager □

Open Device Manager.

- Look for warning symbols or unknown devices.
- Update or reinstall the driver.

Step 4: Verify Driver Compatibility

- Download the latest driver from the manufacturer's website.
- Install the correct version for the operating system.

Step 5: Update Windows □

Install all pending updates.

- Restart the computer.

Step 6: Check for Software Conflicts □

Perform a Clean Boot.

- Disable recently installed software if necessary.

Step 7: Test in Safe Mode

- If the device works in Safe Mode, a third-party application is causing the issue.

Step 8: Verify BIOS/UEFI Settings

- Ensure USB controllers and onboard devices are enabled.

Common Causes

- Corrupted drivers
- Incorrect driver version
- Operating system conflicts
- Faulty USB ports
- Outdated firmware

Conclusion

Following a structured troubleshooting process helps identify hardware and software issues quickly, reducing downtime and ensuring reliable peripheral operation.

Question 4. Explain the role of Uninterruptible Power Supply (UPS) systems and surge protection in preventing file system corruption and ensuring data integrity during unexpected power fluctuations.

Answer. A UPS (Uninterruptible Power Supply) provides temporary battery power during electricity failures, while a surge protector protects electronic devices from sudden voltage spikes. Together, they help protect computer systems, prevent data loss, and improve system reliability.

Role of UPS

A UPS supplies backup power during power outages.

Functions:

- Prevents sudden shutdowns
- Allows users to save work
- Supports safe system shutdown
- Protects hardware during power interruptions

Role of Surge Protection

A surge protector
caused by lightning or electrical faults.

absorbs excess voltage

Benefits:

- Protects motherboard, CPU, RAM, SSD, and PSU
- Reduces hardware damage
- Extends equipment lifespan

Preventing File System Corruption

Unexpected power loss during file operations may corrupt the operating system or stored data.

A UPS helps by:

- Completing active disk operations
- Preventing incomplete file writes
- Reducing operating system corruption
- Protecting important business files

Ensuring Data Integrity

Data integrity means keeping information accurate and complete.

Using a UPS helps:

- Prevent abrupt shutdowns
- Protect databases and documents
- Maintain reliable system performance

Advantages

Device	Benefit
UPS	Prevents data loss during power failure
Surge Protector	Protects hardware from voltage spikes
Battery Backup	Allows safe shutdown
Stable Power	Improves system reliability

Conclusion

UPS systems and surge protectors are essential for protecting computer hardware, preventing file system corruption, maintaining data integrity, and ensuring continuous business operations during unexpected power fluctuations.

SECTION B – Practical Application

Task 1: Physically assemble a workstation and document the component installation sequence.

Objective

To assemble a desktop workstation by installing all hardware components correctly and ensuring the system passes the Power-On Self-Test (POST).

Required Components

- Computer Cabinet (Case)
- Motherboard
- Processor (CPU)
- CPU Cooler
- Thermal Paste
- RAM Module
- NVMe SSD / SATA SSD
- Power Supply Unit (PSU)

- Monitor
- Keyboard
- Mouse
- SATA Cable (if required)
- Power Cable
- Phillips Screwdriver

Workstation Assembly Procedure

Step 1: Prepare the Workspace

- Place the cabinet on a clean, anti-static surface.
- Wear an anti-static wrist strap or discharge static electricity by touching a grounded metal object.

1. Hardware components before assembly



Step 2: Install the CPU

- Open the CPU socket latch on the motherboard.
- Align the golden triangle on the CPU with the socket marker.
- Place the CPU carefully into the socket.
- Lock the socket latch securely.

2. CPU installation



Step 3: Apply Thermal Paste

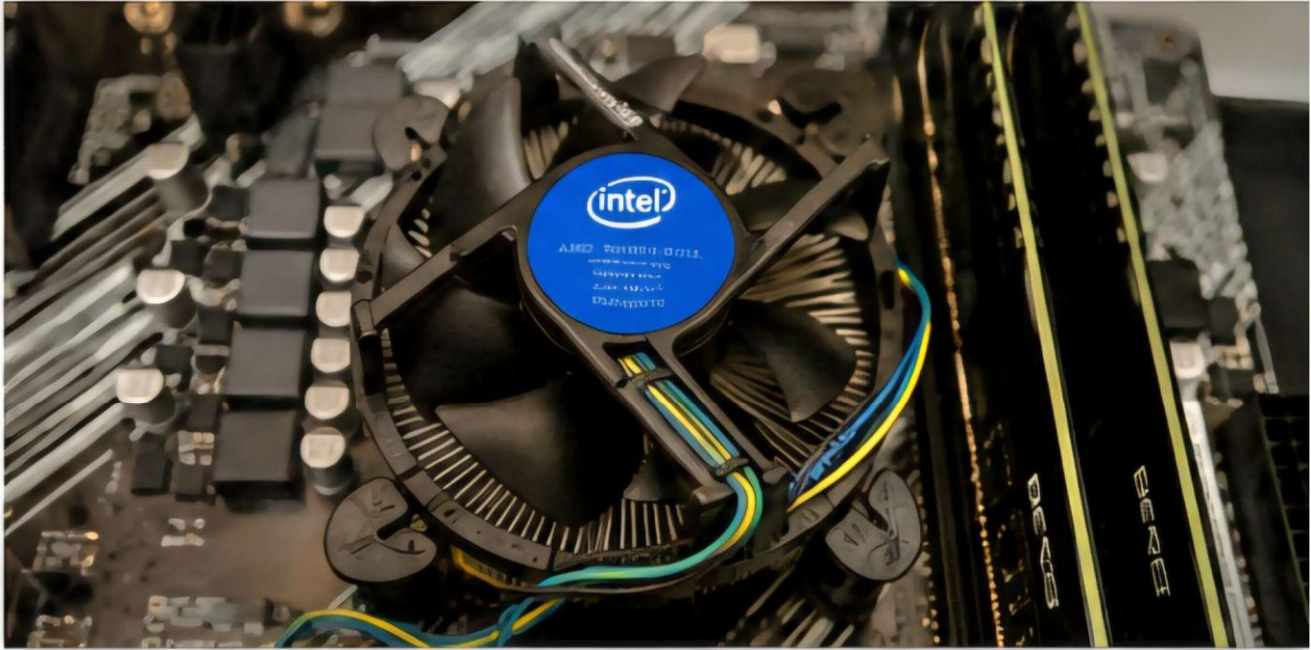
- Apply a small pea-sized amount of thermal paste to the centre of the CPU.
- Spread is not required because the cooler distributes it evenly.

Step 4: Install the CPU Cooler □

Position the cooler over the CPU.

- Tighten the screws diagonally.
- Connect the CPU fan cable to the CPU_FAN header.

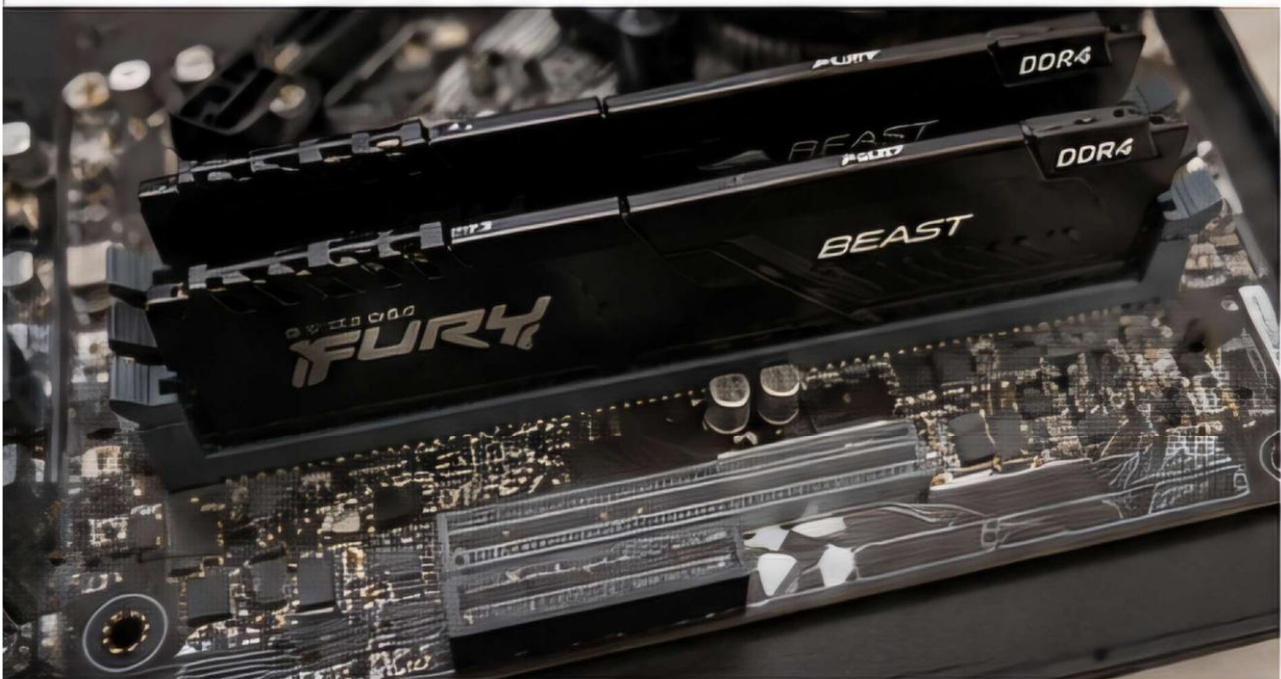
3. CPU cooler installation



Step 5: Install RAM

- Open both RAM slot clips.
- Align the notch on the RAM module.
- Press firmly until both clips lock automatically.

4. RAM installation



Step 6: Install Storage Device

- Insert the NVMe SSD into the M.2 slot at a 30° angle.
- Secure it with the mounting screw.

OR

- Mount the SATA SSD/HDD in the drive bay.
- Connect SATA Data and SATA Power cables.

5. SSD installation



Step 7: Install Motherboard into Cabinet □

Install motherboard standoffs.

- Place the motherboard carefully inside the cabinet.
- Secure it using screws.

Step 8: Install the Power Supply (PSU) □

Position the PSU inside the cabinet.

- Tighten all mounting screws.
- Connect:
 - 24-Pin Motherboard Power Cable
 - 8-Pin CPU Power Cable
 - SATA Power Cable (if required)

6. PSU installation



Step 9: Connect Front Panel Connectors

Connect the following: ☐ Power Switch

- Reset Switch
- HDD LED
- Power LED
- Front USB
- Front Audio

Step 10: Connect External Devices

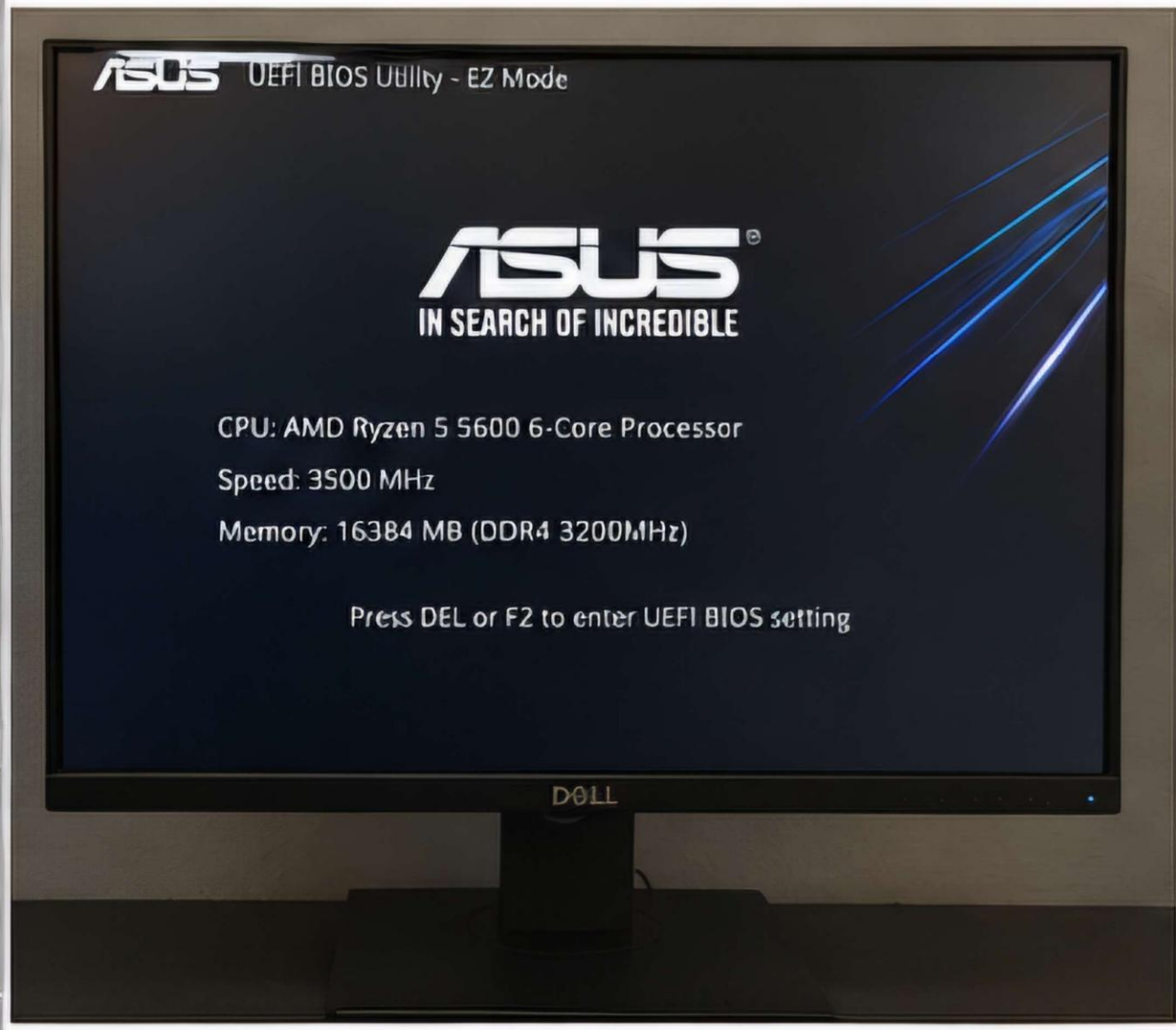
- Monitor
- Keyboard
- Mouse
- Ethernet Cable
- Power Cable

Step 11: Power On the Computer ☐

Switch on the PSU.

- Press the Power Button.
- Verify that the system passes POST successfully.

7. BIOS/POST screen



Result

The workstation was assembled successfully. All hardware components were detected correctly, and the system completed the POST without any errors.

Task 2: Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings.

Objective

To identify and resolve a RAM mismatch error using BIOS/UEFI settings and verify that the installed memory is detected correctly.

Possible Causes of RAM Mismatch ☐

RAM not inserted properly.

- Different RAM capacities.
- Different RAM speeds.
- Incompatible RAM modules.
- Incorrect BIOS settings.
- Faulty RAM slot.
- Outdated BIOS firmware.

Troubleshooting Procedure Step

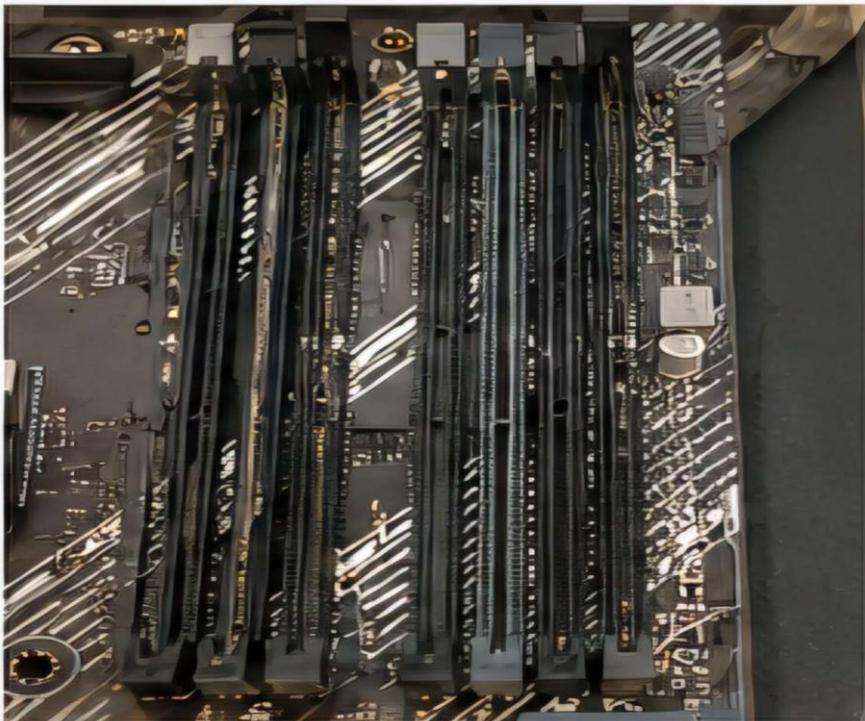
1: Turn O the Computer ☐ Shut down the system.

- Disconnect the power cable.

Step 2: Reseat the RAM

- Remove the RAM module.
- Clean the contacts if necessary.
- Reinsert the RAM firmly until both clips lock.

8. RAM reseated



Step 3: Enter BIOS/UEFI

- Power on the computer.

- Press Delete, F2, or F10 (depending on the motherboard) to enter BIOS.

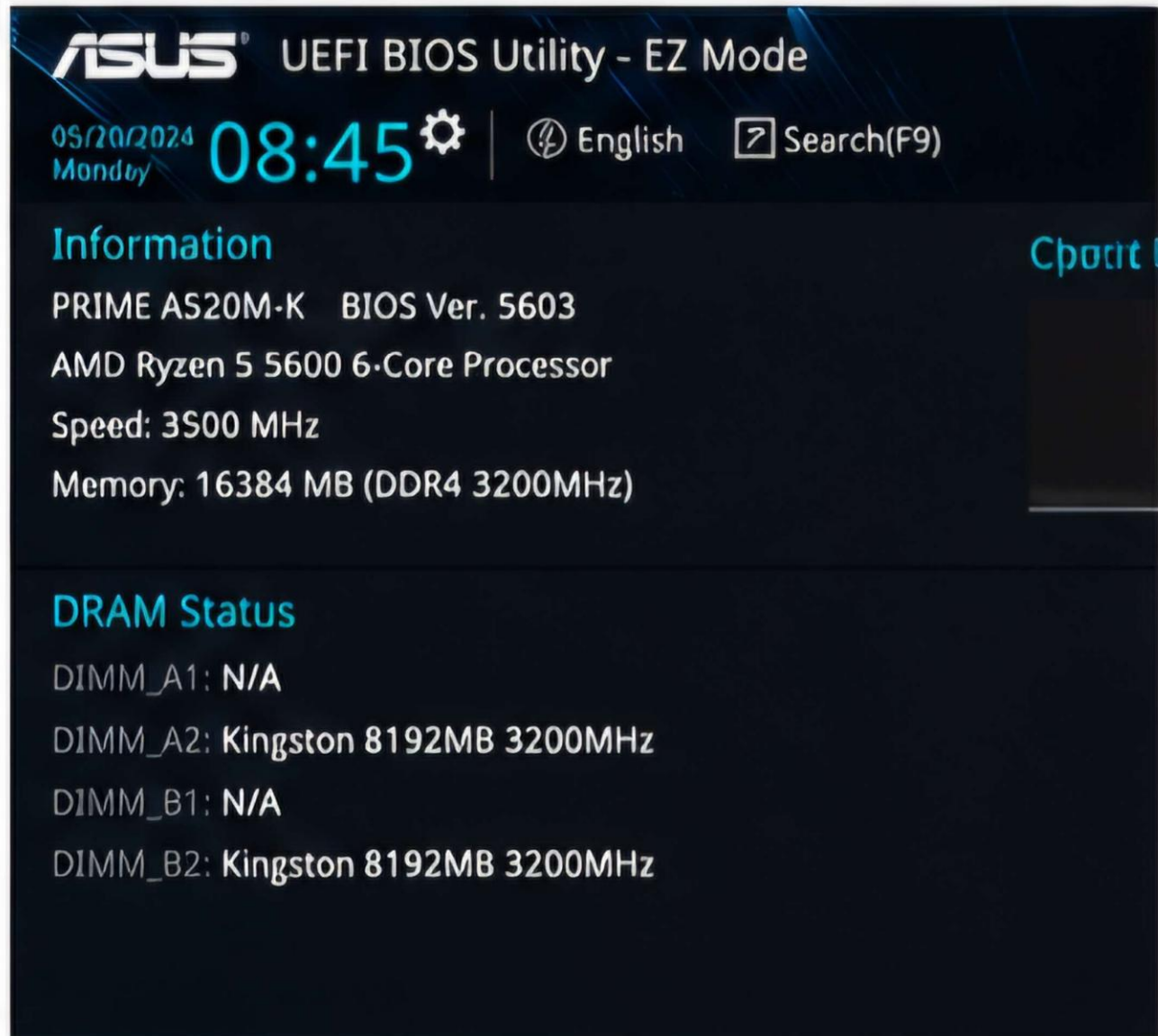
Step 4: Check Installed Memory Navigate to:

Main → System Information

Verify:

- Total Installed Memory
- RAM Frequency
- Number of Memory Slots Used

9. BIOS showing RAM information



Step 5: Enable XMP/DOCP Profile (If supported) ☐
Open Advanced Memory Settings.

- Enable XMP (Intel) or DOCP (AMD).
☐ Save changes.

Step 6: Test Individual RAM Modules ☐

Install one RAM module at a time.

- Test each slot separately.
- Replace any faulty module if required.

Step 7: Update BIOS If

the issue persists:

- Download the latest BIOS from the motherboard manufacturer's website. ☐ Update the BIOS following the official instructions.

Step 8: Save BIOS Settings

- Press F10.
- Save changes.
- Restart the computer. Verification

After restarting:

- Open Task Manager → Performance → Memory.
- Confirm that the correct RAM capacity and speed are displayed.

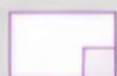
10. Task Manager showing installed RAM

File Options View

Processes Performance App history Startup Users Details Services



CPU
0% .0 0.9 Mbps



Memory
12.6 GB 50 Mbps



Disk 0 (C:)
10%



Wi-Fi
Wi-Fi
2.8 MBps



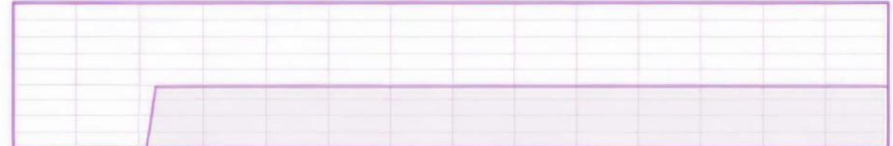
GPU 0
17% 1.7 MBps

Memory

16.0 GB

Memory usage

15.9 GB



80% used

80% used



In use (Compressed)

Available

Speed:

3200 MHz

3.2 GB (0 MB)

12.7 GB

Slots used:

2 of 4

Committed

Cached

Form factor:

DIMM

3.6/18.8 GB

2.1 GB

Hardware reserved:

79.7 MB

Paged pool

Non-paged pool

254 MB

232 MB

Troubleshooting Summary

Problem	Solution
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RAM Not Detected	Reseat the RAM
Di errant RAM Speeds	Enable XMP/DOCP
Faulty RAM	Replace the RAM Module
Faulty Slot	Test Another Slot
BIOS Issue	Update BIOS
Loose Connection	Reinstall RAM Properly

Result

The RAM mismatch issue was successfully resolved by reseating the memory modules, verifying BIOS settings, and confirming the correct memory configuration. The system detected the installed RAM correctly and booted without any errors.

Task 3: Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.

Objective

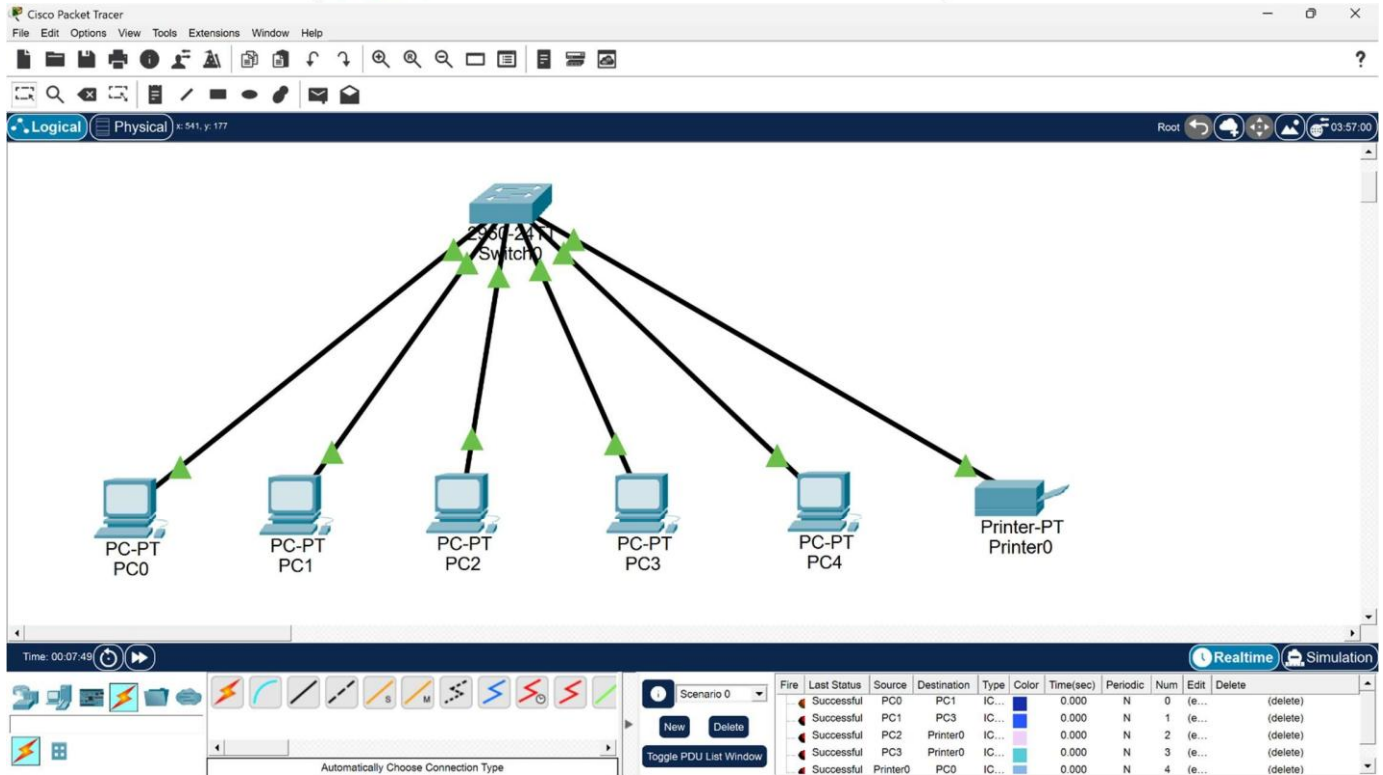
To design and configure a Local Area Network (LAN) using Cisco Packet Tracer for five computers and one printer with a Class C IP addressing scheme.

Software Required

- Cisco Packet Tracer

Devices Required

- 1 Switch (Cisco 2960)
- 5 PCs
- 1 Network Printer
- Copper Straight-Through Cables



Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.2	255.255.255.0	192.168.1.1
PC2	192.168.1.3	255.255.255.0	192.168.1.1
PC3	192.168.1.4	255.255.255.0	192.168.1.1
PC4	192.168.1.5	255.255.255.0	192.168.1.1
PC5	192.168.1.6	255.255.255.0	192.168.1.1
Printer	192.168.1.7	255.255.255.0	192.168.1.1

Configuration Procedure

Step 1: Add Devices

- Open Cisco Packet Tracer.
- Place one Cisco 2960 Switch.
- Add five PCs.
- Add one Printer.

Step 2: Connect Devices

Use Copper Straight-Through Cable to connect:

- PC1 → Switch
- PC2 → Switch
- PC3 → Switch
- PC4 → Switch
- PC5 → Switch
- Printer → Switch

3: Configure IP

Addresses

Open Desktop → IP Configuration on each device and enter the IP addresses according to the table above.

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:2FFF:FE48:E5BA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

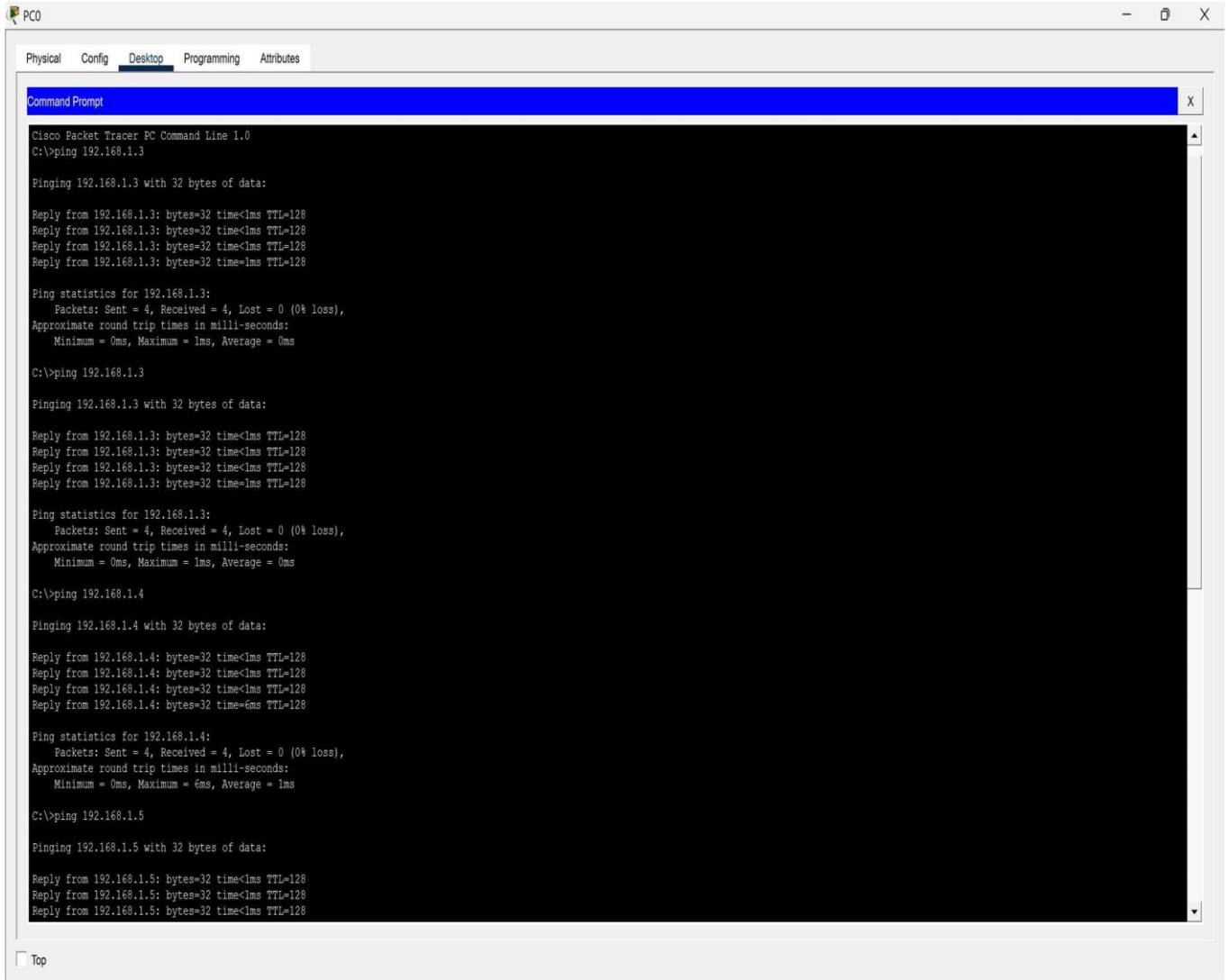
☐ Top

Step 4: Verify Connectivity

Open Command Prompt on PC1 and enter:

ping 192.168.1.3 ping
192.168.1.4 ping
192.168.1.5 ping
192.168.1.6 ping
192.168.1.7

If all replies are successful, the network is configured correctly.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<6ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
```

Result

The LAN topology was successfully created using a Class C subnet. All five computers and the network printer communicated successfully, confirming proper IP configuration and network connectivity.

Task 4: Perform a clean OS installation with manual disk partitioning and startup optimization.

Objective

To install a clean operating system using manual disk partitioning and optimize startup programs for better system performance.

Software Required

- Windows 10/11 ISO
- Bootable USB Drive
- VirtualBox or Physical Computer

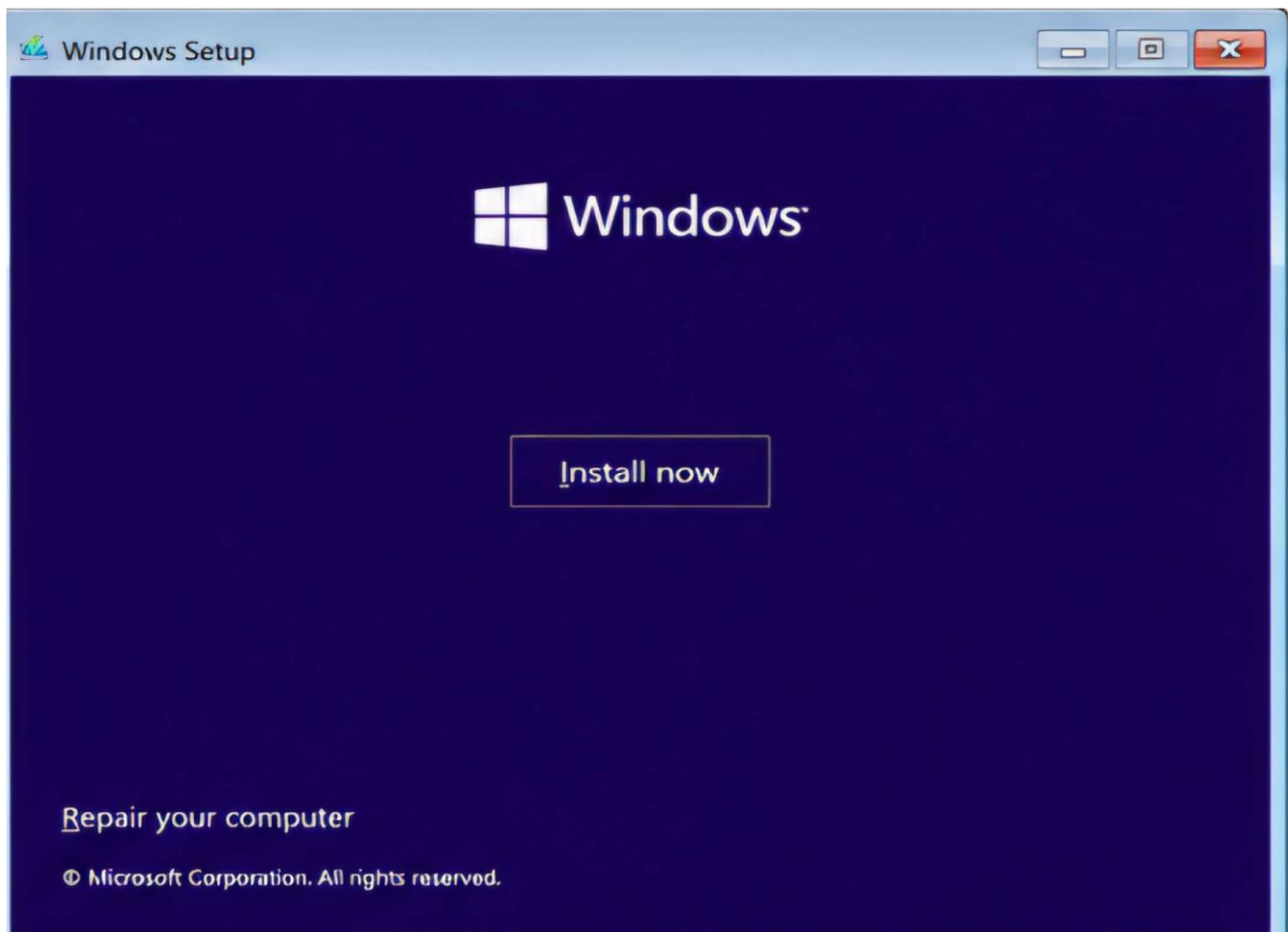
Installation Procedure

Step 1: Create Bootable USB

- Download the Windows ISO.
- Create a bootable USB using Rufus or the Media Creation Tool.

Step 2: Boot from USB

- Restart the computer.
- Press the Boot Menu key (F12, Esc, or F9 depending on the system). ☐ Select the USB drive.



Step 3: Begin Installation ☐

- Select Language.
- Select Time and Currency Format.
- Select Keyboard Layout.

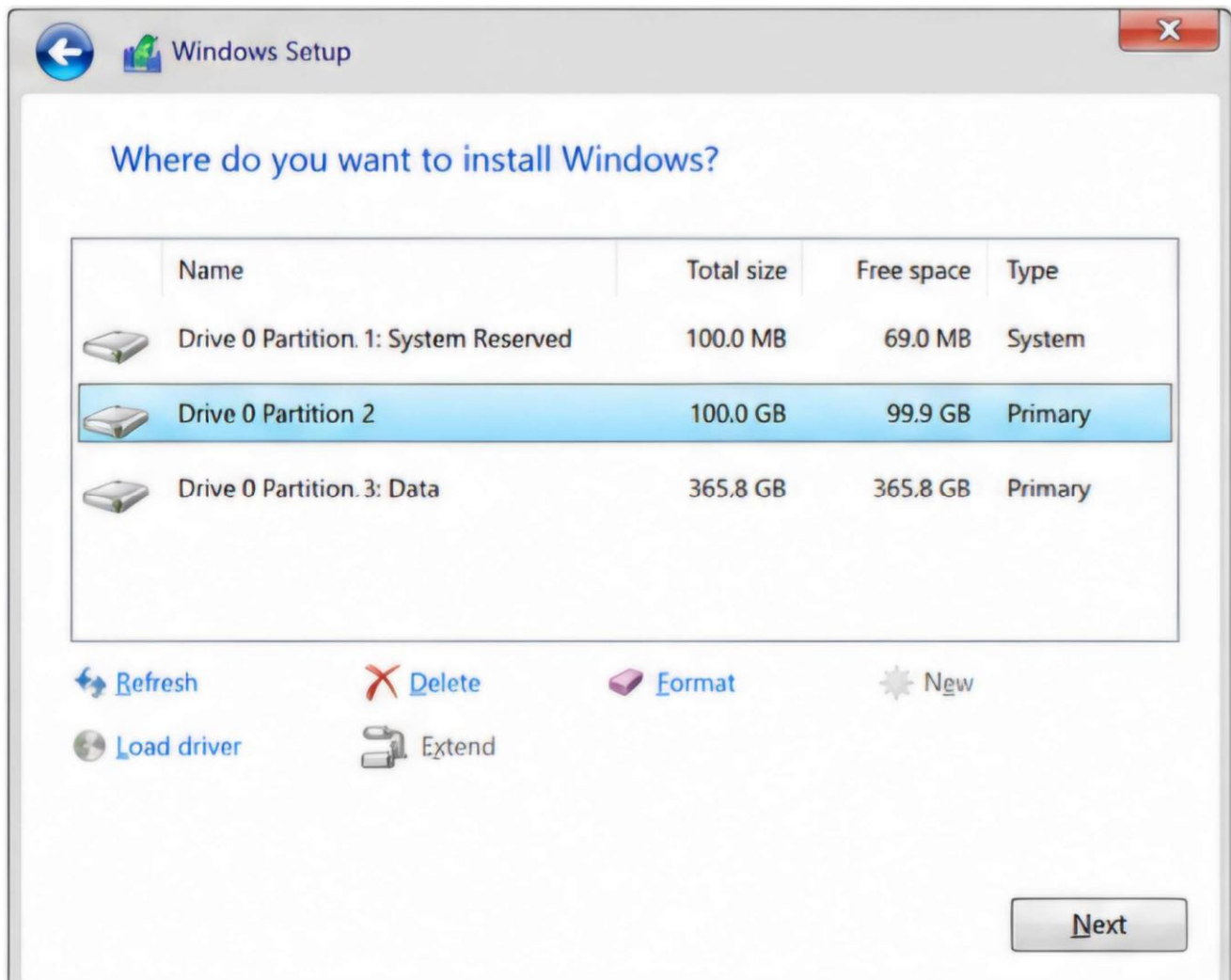
- Click Install

Now.

Step 4: Manual Disk Partitioning Choose Custom: Install Windows Only. Create the following partitions:

Partition	Size
System (C:)	100 GB
Data (D:)	Remaining Space

Select the C: partition and click Next.



Step 5: Complete Installation

- Wait for Windows to copy files.
- Restart the computer.
- Complete the initial Windows setup by creating a user account and setting preferences.



Startup Optimization

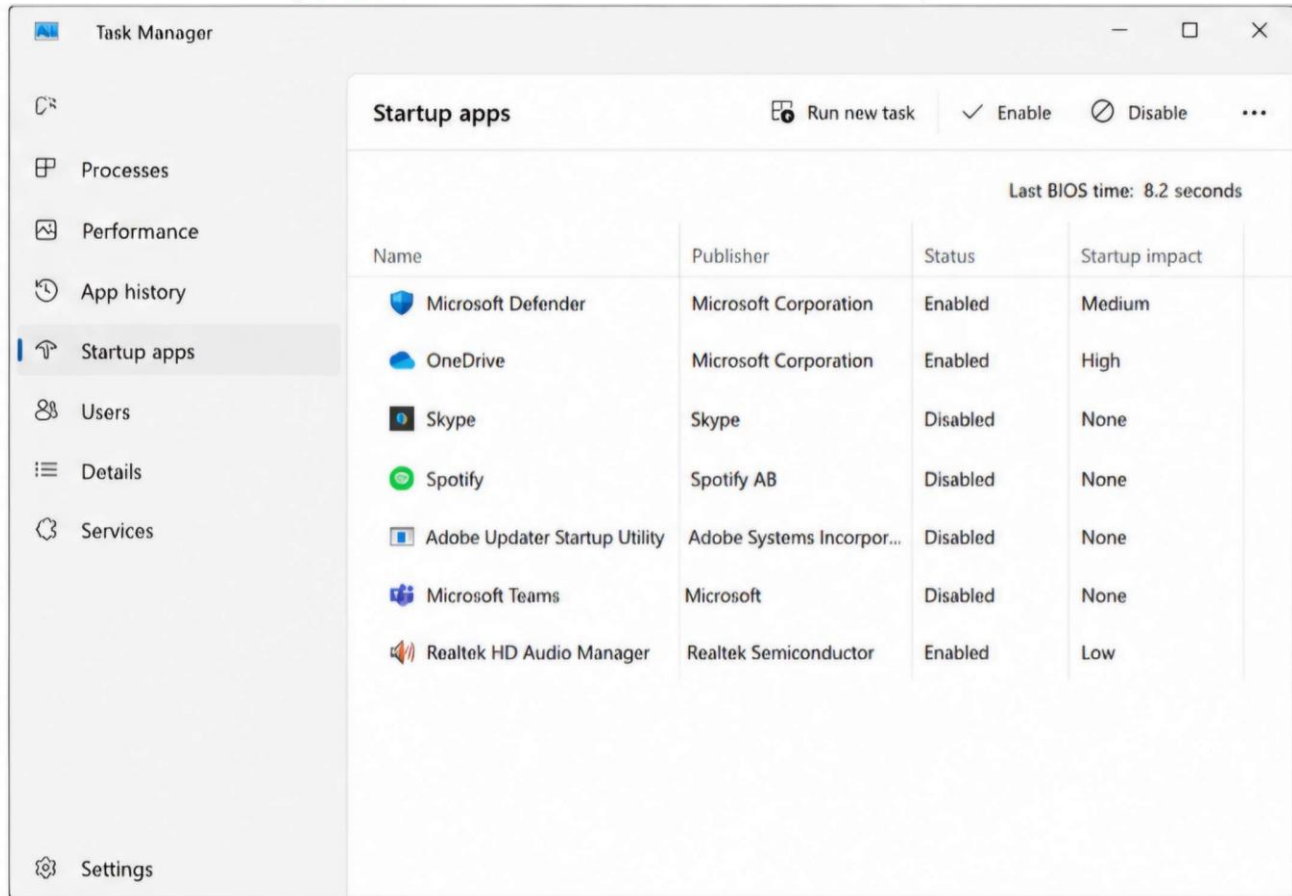
Step 1: Open Task Manager

- Press Ctrl + Shift + Esc.
- Go to the Startup Apps tab.

Step 2: Disable Unnecessary Startup Programs

Disable applications that are not required at startup, such as:

- Skype
- Discord
- Spotify
- Adobe Updater
- Microsoft Teams (if not used) Keep essential applications enabled:
- Windows Security
- Audio Drivers
- Graphics Drivers



Step 3: Install Drivers Install

the following drivers: ☐

Chipset Driver

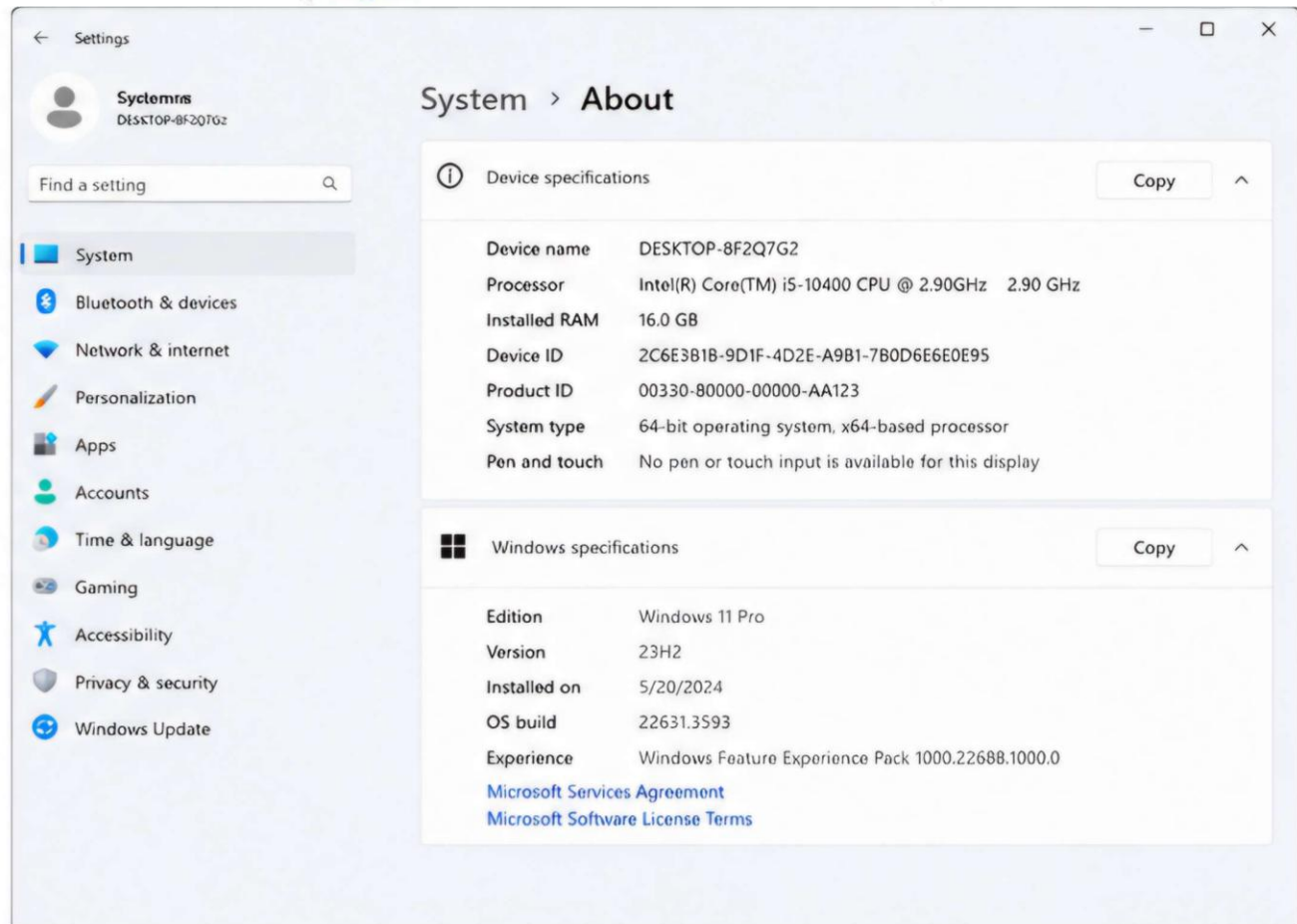
- Graphics Driver
- Network Driver
- Audio Driver

Step 4: Install Windows Updates

- Open Settings → Windows Update.
- Download and install all available updates.

Step 5: Verify Installation Check:

- Windows activates successfully.
- All hardware devices are detected.
- No unknown devices appear in Device Manager.
- System boots without errors.



Result

The operating system was successfully installed using manual disk partitioning. Startup applications were optimized to improve boot time and system performance. All required drivers were installed, Windows updates were completed, and the system operated without errors.

SECTION C – Integrated Case Study

Scenario

At Nexus Innovations, three rendering workstations pass morning POST checks but restart randomly during heavy afternoon workloads. The units are stored in poorly ventilated desk cabinets.

Question 1. Identify the three hardware components most likely to cause random reboots under load.

Answer. The three hardware components most likely responsible for random reboots are:

1. CPU (Processor)

The CPU generates significant heat during rendering tasks. If it overheats due to poor cooling, the system may automatically restart to prevent damage.

2. Power Supply Unit (PSU)

A faulty or low-wattage PSU may fail to provide stable power during heavy workloads, causing unexpected restarts.

3. CPU Cooling System

A malfunctioning CPU fan, clogged heatsink, or dried thermal paste can cause excessive temperatures, resulting in automatic shutdowns or restarts.

Component	Possible Issue
CPU	Overheating under heavy load
PSU	Insufficient or unstable power
CPU Cooler	Poor cooling or fan failure

Conclusion

The CPU, PSU, and cooling system are the most likely hardware components responsible for random restarts during intensive workloads.

Question 2. Explain why the systems pass POST in the morning but fail in the afternoon.

Answer. The systems successfully pass the Power-On Self-Test (POST) in the morning because the hardware is cool and operating within normal temperature limits.

As the day progresses:

- Continuous rendering increases CPU and GPU temperature.
- Poor ventilation inside the desk cabinets traps hot air.
- Cooling efficiency decreases.
- The CPU reaches its thermal limit.
- The system automatically restarts to protect the hardware.

Reasons

- High room temperature.
- Poor airflow inside the cabinet.
- Continuous heavy workload.

- Dust fans. accumulation in cooling
- Thermal throttling due to overheating.

Conclusion

The systems operate normally when cool but restart later because heat builds up during prolonged workloads, causing thermal protection mechanisms to activate.

Question 3. How would you use a stress-test tool to confirm a thermal or power issue? Answer. A stress-test tool is used to place the CPU and other components under maximum load while monitoring system temperature and stability.

Procedure

1. Install a stress-testing tool such as Prime95, AIDA64, or OCCT.
2. Open a hardware monitoring tool to observe CPU temperature and power usage.
3. Start the stress test for 15–30 minutes.
4. Monitor:
 - o CPU temperature
 - o Fan speed
 - o System stability
 - o Voltage readings
5. If the system restarts, freezes, or temperatures exceed safe limits, the issue is likely related to overheating or insufficient power.

Expected Result

A stable system should complete the test without restarting or exceeding safe operating temperatures.

Conclusion

Stress testing helps identify thermal and power-related issues by simulating heavy workloads and monitoring system behaviour.

Question 4. Propose three specific steps to remediate the frequent system restarts. Answer. The following steps can help resolve the frequent restart issue:

1. Improve Cooling

- Clean dust from fans and heatsinks.
- Replace old thermal paste.
- Ensure all cooling fans are functioning properly.

2. Improve Airflow

- Move the workstations out of enclosed desk cabinets.
- Provide adequate ventilation around each system. □ Maintain a cool room temperature.

3. Check the Power Supply

- Verify that the PSU provides sufficient wattage.
- Replace a faulty or unstable PSU if required.
- Ensure all power cables are securely connected.

Conclusion

Improving cooling, increasing airflow, and verifying power supply stability will significantly reduce random restarts.

Question 5. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

Answer. A PSU upgrade is necessary only if the existing power supply cannot deliver sufficient and stable power to all hardware components during heavy workloads.

A PSU may be required when:

- The installed PSU has low wattage.
- Voltage fluctuates under load.
- A dedicated graphics card or additional hardware has been installed. ☐ The current PSU is old or faulty.

A PSU may not be required if:

- The existing PSU has adequate wattage.
- Voltage output remains stable during stress testing.
- The restart issue is caused only by overheating or poor ventilation.

Conclusion

A PSU upgrade should be performed only after testing confirms that the current power supply is inadequate or unstable. If overheating is the actual cause, improving cooling and airflow will solve the problem without replacing the PSU.

Overall Case Study Conclusion

The random restarts are primarily caused by overheating due to poor ventilation and continuous heavy workloads. Proper cooling, better airflow, routine maintenance, and verification of PSU performance will improve system stability and prevent unexpected restarts.